

Differential geometry

Part III

- manifolds, tangent spaces, retractions
- Lie groups $SO(N)$, Lie algebras $\mathfrak{g}$
- homogeneous spaces $St(n, N)$, $Gr(n, N)$
- $\mathfrak{g} = \mathfrak{g}^{\text{ver}} \oplus \mathfrak{g}^{\text{hor}}$, global sections $\lambda(Y)$
- geodesics, parallel transport

Numerics

Parts I & III

- function bases: splines per patch, degree p
- FEEC: a *subcomplex*, commuting π_h
- exp: Taylor and Padé kernels
- scaling & squaring, Cayley, QR completion
- backward error analysis

`GeometricOptimizers.jl`

Geometric machine learning

Part II

- SympNets
- symplectic autoencoders

Structure preservation

Parts I & II

a symplectic form

- ω , $\dot{z} = \mathbb{J} \nabla_z H$, symplectic flows
- Poincaré invariants, non-squeezing
- symplectic integrators, modified H_h

a de Rham complex

- $d \circ d = 0$, so $\text{im } d \subseteq \ker d$
- cohomology $\dim \mathcal{H}^k = b_k(\Omega)$
- harmonic fields
 $\ker \text{curl}_h / \text{im } \text{grad}_h$

a symmetry

- Noether, discrete momentum maps